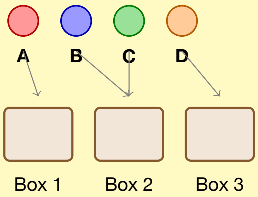
DISTRIBUTING OBJECTS INTO BOXES

1. DISTINGUISHABLE OBJECTS & DISTINGUISHABLE BOXES

Objects are **different**.
Boxes are **different** (labeled).



Interpretation:
Each object can go to any box.

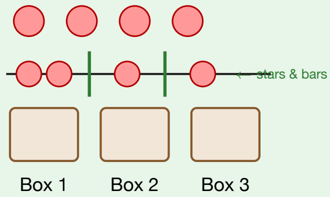
Total ways:
n^m

n = number of boxes
m = number of objects

Example: $3^4 = 81$ ways

2. INDISTINGUISHABLE & DISTINGUISHABLE

Objects are **identical**.
Boxes are **different** (labeled).



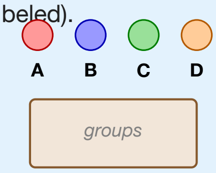
Interp
Non-
$x_1 +$

n = nu
m = nu

Example: $\binom{4+3-1}{3-1} = \binom{6}{2} = 15$ ways

3. DISTINGUISHABLE OBJECTS & INDISTINGUISHABLE BOXES

Objects are **different**.
Boxes are **identical** (unla-beled).



Interpretation:
Partitioning objects into groups.

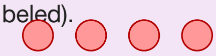
Total ways:
Bell number
B_m

Into exactly k groups:
Stirling number $S(m, k)$
$B_m = \sum_{k=0}^{m} S(m, k)$

Example: $m = 4$: $B_4 = 15$ ways (Bell partitions of 4 objects)

4. INDISTINGUISHABLE & INDISTINGUISHABL

Objects are **identical**.
Boxes are **identical** (unla-beled).



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Partitions of 4:

4
$3 + 1$
$2 + 2$
$2 + 1 + 1$
$1 + 1 + 1 + 1$

$p(m)$ =
sum of
(order

Example: $m = 4$: $p(4) = 5$ ways

Quick Reference — GATE DA

headerbg!12	Distinguishable Boxes	Indistinguishable Boxes
Distinguishable Objects	n^m	Bell number B_m
Indistinguishable Objects	$\binom{m+n-1}{n-1}$ (Stars & Bars)	Partitions $p(m)$

m = objects, n = boxes | **Most common in GATE:** Cases 1 & 2